Andrés Felipe Joya Buitrago

Systems Engineering Student | Python, Laravel, React, SQL and NoSQL

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SUMMARY

Systems Engineering student (Universidad Autónoma de Bucaramanga, 2023–present) focused on software development and relational and NoSQL databases (MySQL, MongoDB, Oracle APEX). I build applications in Python, PHP (Laravel), TypeScript and Kotlin, and hold AWS and MongoDB certifications. Proactive and results-oriented: I aim to optimize processes and contribute to technology and administrative projects.

EXPERIENCE

Logistics Volunteer | Ulibro (university events volunteering) | August 2025

- Supported the logistics and organization of several events and managed the organization of one section of the event, “Amigos del Agua”.

PROJECTS

- **Inventory Manager** (PHP, Laravel 10, Livewire, Jetstream, Docker): web application to manage products, customers, suppliers and stock movements; consistency guaranteed with transactions and row locking, non-negative stock validation and edge-case tests.
- **MoodJournal** (React, TypeScript, Supabase, Tailwind CSS): mood journal with in-browser ES/EN sentiment analysis, statistics, writing streaks, authentication and row-level security (RLS); CI and Vercel deployment.
- **Genetic algorithm for investment portfolio** (Python, NumPy, Tkinter, JavaScript): optimizes the return/risk ratio of 10 assets (roulette selection, crossover with repair, mutation and elitism); desktop and web versions, with pytest and node tests.
- **BUAP School Manager** (Laravel 12, Tailwind CSS, PostgreSQL, Docker): MVC academic system to manage subjects, groups, students and enrollments, with authentication (Breeze) and Pest tests.
- **StockWise** (FastAPI, SQLAlchemy, PostgreSQL, Docker Compose): inventory management backend, in development.

SKILLS

Languages: Python, Java, Kotlin, PHP, TypeScript, JavaScript

Backend and web: Laravel, Flask, FastAPI, React, HTML, CSS, Tailwind CSS

Mobile: Kotlin and Android Studio

Databases: MySQL, PostgreSQL, MongoDB, Oracle, Oracle APEX, Supabase (SQL and NoSQL)

Cloud and infrastructure: AWS, Microsoft Azure (IoT Central), Docker, networking and infrastructure

Analysis and design: Data modeling; UML, flowcharts, use cases and sequence diagrams

Tools: Git and GitHub, Excel, Word, PowerPoint

Soft skills: teamwork, assertive communication, leadership, working under pressure, problem solving, adaptability and continuous learning, analytical thinking, resource and process management, decision making

EDUCATION

Systems Engineering, Universidad Autónoma de Bucaramanga | 2023 – present

High School (Bachillerato), Liceo San Fernando | 2020 – 2022

Primary School, Colegio Niño Jesús de Praga | 2014 – 2019

CERTIFICATIONS

Introduction to MongoDB | MongoDB | 2025

Networking Academy Learn-A-Thon | Cisco | 2025

AWS Academy Graduate – AWS Academy Cloud Foundations | Amazon Web Services (AWS) | 2025

AWS Academy Graduate – AWS Academy Data Engineering | Amazon Web Services (AWS) | 2025

AWS Academy Graduate – Generative AI Foundations (Training Badge) | Amazon Web Services (AWS) | 2025

AWS Academy Graduate – Machine Learning for Natural Language Processing | Amazon Web Services (AWS) | 2025

LANGUAGES

Spanish (native) | English (C1) | Portuguese (basic)

PROJECTS (DETAIL)

Inventory Manager | PHP, Laravel 10, Jetstream, Livewire 3, Tailwind CSS, Docker | Web application

- Management of products, customers, suppliers and stock movements (inbound and outbound) for small warehouses, with a KPI dashboard, low-stock alerts and a trash bin with restore.
- Stock consistency through a service layer with transactions and row locking (*lockForUpdate*); dedicated exception when an operation would leave negative stock; automatic recalculation when movements are edited or deleted.
- Full authentication with Jetstream (password recovery, two-factor); edge-case tests, CI and deployment prepared with Dockerfile and Render.
- Code: github.com/Sandrez12/Gestor-de-Inventario

MoodJournal | React, TypeScript, Vite, Tailwind CSS, Supabase (Auth and PostgreSQL) | Web application

- Mood journal with date-based entries, statistics, emotional chart, writing streaks, per-user preferences and light/dark mode.
- Lexicon-based sentiment analysis (ES/EN) with negation handling, run in the browser (text never leaves the client) and covered by tests; row-level security (RLS) to isolate each user's data.
- CI and Vercel deployment. Demo: moodjournal-theta.vercel.app
- Code: github.com/Sandrez12/MoodJournal

Genetic algorithm for investment portfolio | Python, NumPy, Tkinter, Matplotlib, JavaScript | Optimization

- Genetic algorithm that searches for the allocation of 10 assets with the best return/risk ratio (similar to the Sharpe ratio) from the mean vector and covariance matrix.
- Roulette selection, one-point crossover with largest-remainder repair (individuals are always valid), configurable mutation and elitism; Python core with reproducible seed.
- Desktop interface (Tkinter) and interactive dependency-free web version; tests with pytest and node test; CI and demo on GitHub Pages.
- Code: github.com/Sandrez12/Algoritmo-gen-tico-aplicado-a-un-portafolio-de-inversi-n

BUAP School Manager | PHP, Laravel 12, Breeze, Tailwind CSS, PostgreSQL/MySQL, Docker | Web application

- Academic system (MVC) to manage subjects, groups, students and enrollments, with an academic workload dashboard and authentication (registration, password recovery and profile).
- Pest tests and a Docker image (PHP + Apache) ready for Render.
- Code: github.com/Sandrez12/GestorEscolarBUAP

StockWise | Python, FastAPI, SQLAlchemy 2, PostgreSQL 16, Docker Compose | In development

- Inventory management backend: base API and PostgreSQL connection configured through environment variables; in progress: models, CRUD endpoints, authentication and tests.
- Code: github.com/Sandrez12/StockWise

Good practices concepts | PHP, Laravel 12, Docker | Educational site

- Site with before/after examples on SOLID principles, cohesion and coupling, and dependency injection, applying what it teaches (content decoupled through an injected interface); accessible and with CI.
- Code: github.com/Sandrez12/Conceptos-

Bilingual personal portfolio | PHP, Laravel 12, Tailwind CSS, Docker | Website

- ES/EN site with projects, skills and a contact form with validation, rate limiting, anti-bot field and reply-to email; SEO/Open Graph metadata and accessibility.
- Code: github.com/Sandrez12/Portafolio-Andres-Felipe-Joya-Buitrago